「Orthogonal Projection - Revised.

Let V be an inn. prod space, (not necessary finite dimensional) and $W \leq V$ be a subspace.
For any vector $\beta \in V$, if $\alpha \in W$ satisfies that:

$$\|\beta - \alpha\| \leq \|\beta - \gamma\| \quad \text{for all } \gamma \in W \quad \cdots (*)$$

Then, the α is called best approximation to β by W, simply b.app.

「Thm. $\alpha \in W$ is b.app to $\beta \in V$ by W iff for all $\mu \in W$, $\langle \beta - \alpha, \mu \rangle = 0$.

Proof. For fixed $\alpha \in W$ and $\beta \in V$, for any $\gamma \in W$,

$$\|\beta - \gamma\|^2 = \|\beta - \alpha + \alpha - \gamma\|^2 = \|\beta - \alpha\|^2 + \|\alpha - \gamma\|^2 + 2 \cdot \mathrm{Re}\langle \beta - \alpha, \alpha - \gamma \rangle$$

Therefore, the condition $(*)$ holds if and only if

$$0 \leq \|\alpha - \gamma\|^2 + 2 \cdot \mathrm{Re}\langle \beta - \alpha, \alpha - \gamma \rangle \quad \text{for all } \gamma \in W.$$

Now, take $\gamma = \alpha + \dfrac{\langle \beta - \alpha, \alpha - \mu \rangle}{\|\alpha - \mu\|^2} \cdot (\alpha - \mu)$ for any $\mu \in W \setminus \{\alpha\}$.

Then, $$0 \leq \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} - 2 \cdot \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} = - \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2}$$

And the inequality holds iff $\langle \beta - \alpha, \alpha - \mu \rangle = 0$, and all nonzero vectors in W can be represented by $\alpha - \mu$, thus proved.

Conversely, if $\beta - \alpha$ is orthogonal to every vector in W, then $\mathrm{Re}\langle \beta - \alpha, \alpha - \gamma \rangle = 0$, thus $\|\beta - \gamma\|^2 - \|\beta - \alpha\|^2 = \|\alpha - \gamma\|^2 \geq 0$. ☒

「Thm. If b.app to $\beta \in V$ by W exists, then it is unique.

Proof. Suppose that $\alpha_1, \alpha_2 \in W$ satisfies for all $\gamma \in W$,

$$\langle \beta - \alpha_1, \gamma \rangle = \langle \beta - \alpha_2, \gamma \rangle = 0.$$

Then, take $\gamma = \alpha_1 - \alpha_2$, so $\langle \alpha_1 - \alpha_2, \alpha_1 - \alpha_2 \rangle = 0$. ☒

「Def. For all $\beta \in V$, there is b.app $\alpha \in W$, then the map

$$P : V \to V : \beta \mapsto \alpha$$

is called an orthogonal projection on W.

「Thm. Let W be a fin. dim subspace of inn. prod space V.
Then, the unique orthogonal projection on W exists. Further,
P is idempotent linear operator with $\mathrm{Im}\, P = W$, $\ker P = W^\perp$.

Proof. Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis for W. Then, a map

$$P : V \to V : \beta \mapsto \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \cdot \alpha_i$$

$\frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i$ $\quad$ $\frac{\langle \gamma, \alpha_i \rangle}{\|\alpha_i\|^2}$.

1). It maps β to orthogonal projection by W.
For each $1 \le j \le n$,

$$\left\langle \beta - \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \alpha_i, \alpha_j \right\rangle = \langle \beta, \alpha_j \rangle - \langle \beta, \alpha_j \rangle = 0.$$

So, $\beta - P(\beta) \in W^\perp$ for all $\beta \in V$. (And, such a vector is unique, thus well-defined.)

2). It is linear.
Let $\beta, \gamma \in V$ and $c \in F$ be given. Then,

$$P(c\beta + \gamma) = \sum \langle c\beta + \gamma, \alpha_i \rangle \cdot \alpha_i = c \cdot \sum \langle \beta, \alpha_i \rangle \cdot \alpha_i + \sum \langle \gamma, \alpha_i \rangle \cdot \alpha_i = cP(\beta) + P(\gamma).$$

3). It is idempotent.
Given $\beta \in V$, $P(\beta) \in W$. So, $P^2(\beta) = P(\beta)$. $\perp$ projection

4). $\mathrm{Im}\, P = W$, $\ker P = W^\perp$.
$\mathrm{Im}\, P \subseteq W$ is clear, and for any $\gamma \in W$, $P(\gamma) = \gamma \in \mathrm{Im}\, P$. And,

$$\beta \in \ker P \iff P(\beta) = 0 \iff \langle \beta, \alpha_i \rangle = 0 \text{ for all } 1 \le i \le n \iff \beta \in W^\perp.$$

5). It is Hermitian.
Since $\mathrm{Im}\, P \oplus \ker P = V$, let $x_1, y_1 \in \mathrm{Im}\, P$ and $x_2, y_2 \in \ker P$ be given. Then

$$\langle P(x_1 + x_2), y_1 + y_2 \rangle = \langle x_1, y_1 + y_2 \rangle = \langle x_1, y_1 \rangle = \langle x_1 + x_2, y_1 \rangle = \langle x_1 + x_2, P(y_1 + y_2) \rangle.$$

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「Corollary. Let W be a fin. dim subspace of inn. prod space V,
and let P be an orthogonal projection on W. Then, $I - P$ is an orthogonal projection on $W^\perp$.

Proof. Given $\beta \in V$, $\langle \beta - P(\beta), \mu \rangle = 0$ for all $\mu \in W$, or $\beta - P(\beta) \in W^\perp$. So $\mathrm{Im}(I - P) = W^\perp$.
And, given $\gamma \in W^\perp$, $\beta - \gamma = \underbrace{P(\beta)}_{W} + \underbrace{(\beta - P(\beta) - \gamma)}_{\in W^\perp}$, therefore

$$\|\beta - \gamma\|^2 = \|P(\beta)\|^2 + \|\beta - P(\beta) - \gamma\|^2 \ge \|P(\beta)\|^2 = \|\beta - (\beta - P(\beta))\|^2 \text{ for all } W^\perp,$$

So $\beta - P(\beta)$ is the best approximation to β by vectors in $W^\perp$.

Note: let $E: V \to V$ be a projection. In general, E can not be determined by it's image.
For example, let $V = \mathbb{R}^3$, $W_1 = \text{Span}\{(1,0,0)\}$, $W_2 = \text{span}\{(0,1,0), (0,0,1)\}$, $W_3 = \text{Span}\{(0,1,0), (1,1,1)\}$.
Then, $V = W_1 \oplus W_2 = W_1 \oplus W_3$, but $W_2 \neq W_3$. Moreover,
$E: V \to V : (x_1, x_2, x_3) \mapsto (x_1, 0, 0)$, $E': V \to V ; (x_1, x_2, x_3) \mapsto (x_1 - x_3, 0, 0)$
Both E and E' are projections with $\text{Im}\,E = \text{Im}\,E' = \text{Span}\{(1,0,0)\}$ but $E = E'$,
$\ker E = W_2$ and $\ker E' = W_3$.

Corollary. Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthogonal set of non-zero vectors in inn. prod space V. Then,
for all $\beta \in V$, $\displaystyle\sum_{i=1}^{n} \frac{|\langle \beta, \alpha_i \rangle|^2}{\|\alpha_i\|^2} \leq \|\beta\|^2$; the equality holds iff $\displaystyle\beta = \sum \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i$.

pf. Take $\gamma = \sum \left[\langle \beta, \alpha_i \rangle / \|\alpha_i\|\right] \cdot \alpha_i$. Then, $\langle \beta - \gamma, \gamma \rangle = 0$. So,

$$\|\beta\|^2 = \|\gamma\|^2 + \|\beta - \gamma\|^2 \geq \|\gamma\|^2.$$

Thm. Let W be a finite-dimensional subspace of an inn. prod space V.
If $P: V \to V$ is an orthogonal projection on W, that is,

$$\text{for any } \beta \in V, \quad \|\beta - P(\beta)\| \le \|\beta - \gamma\| \text{ for all } \gamma \in W \quad \cdots (*)$$

then, P is idempotent hermitian linear map s.t. $\operatorname{Im} P = W$, $\ker P = (\operatorname{Im} P)^{\perp}$, $V = \operatorname{Im} P \oplus \ker P$

Proof. Since $P(\beta) \in W$, $P(P(\beta)) = P(\beta)$, for all $\beta \in V$, so $P^2 = P$, idempotent.
To show the linearity, let $\alpha, \beta \in V$ and $c \in F$ be given. Since $\alpha - P(\alpha)$, $\beta - P(\beta) \in W^{\perp}$, so

$$c \cdot (\alpha - P(\alpha)) + (\beta - P(\beta)) = (c\alpha + \beta) - (c \cdot P(\alpha) + P(\beta)) \in W^{\perp}.$$

So, $cP(\alpha) + P(\beta) = P(c\alpha + \beta)$, linearity is obtained
To show the direct sum, given $\beta \in V$, $\beta = P(\beta) + (\beta - P(\beta))$, and since $\beta - P(\beta) \in W^{\perp}$,
$V = \operatorname{Im} P + W^{\perp}$ is obtained. And,

$$\alpha \in W^{\perp} \iff P(\alpha) = 0, \text{ because}$$

$\alpha \in W^{\perp} \iff \forall \mu \in W,\ \underset{\langle \alpha - 0, \mu \rangle = 0}{\langle \alpha, \mu \rangle \overset{\shortparallel}{=} 0} \iff P(\alpha) = 0$. So, $V = \operatorname{Im} P + \ker P$. And, $\operatorname{Im} P \cap \ker P = \{0\}$,

because $P(\alpha) = 0$ and $\alpha = P(\alpha)$ implies $\alpha = 0$.